

Heart Disease Prediction

A Hybrid Machine Learning Approach: Comparing Logistic Regression, Random Forest, XGBoost & Deep Learning

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Problem & Motivation

Heart disease remains the leading cause of mortality worldwide, claiming millions of lives annually. Despite its prevalence, many cases go undetected until it is too late for effective intervention.

Early prediction enables preventive care and significantly improves patient outcomes. Machine learning empowers clinicians with predictive analytics that support faster, data-driven decisions.

ML models enhance risk stratification, helping healthcare systems allocate resources more efficiently and prioritize high-risk patients before critical events occur.

Harnessing the power of AI in cardiology can transform reactive treatment into proactive, life-saving prevention.



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Dataset Overview

918 patient records with 11 clinical features, providing a well-balanced dataset for binary heart disease classification.

Key Features: Age, Sex, Chest Pain Type, Resting BP, Cholesterol, Fasting Blood Sugar, Resting ECG, Max Heart Rate, Exercise-Induced Angina, Oldpeak, ST Slope.

Target Variable: HeartDisease — 508 positive cases, 410 negative cases. Well-balanced for classification tasks.

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Aosomrod	1	♥	♥	2 oom	—		
Pactrandan	1	♥	♥	1 oom	—		
Harkdaffle	1	♥	♥	2 oom	—		
Denzatic	1	♥	♥	2 oom	—		
Plemael		♥	♥	1 oom	—		
Dommel		♥	♥	2 oom	—		
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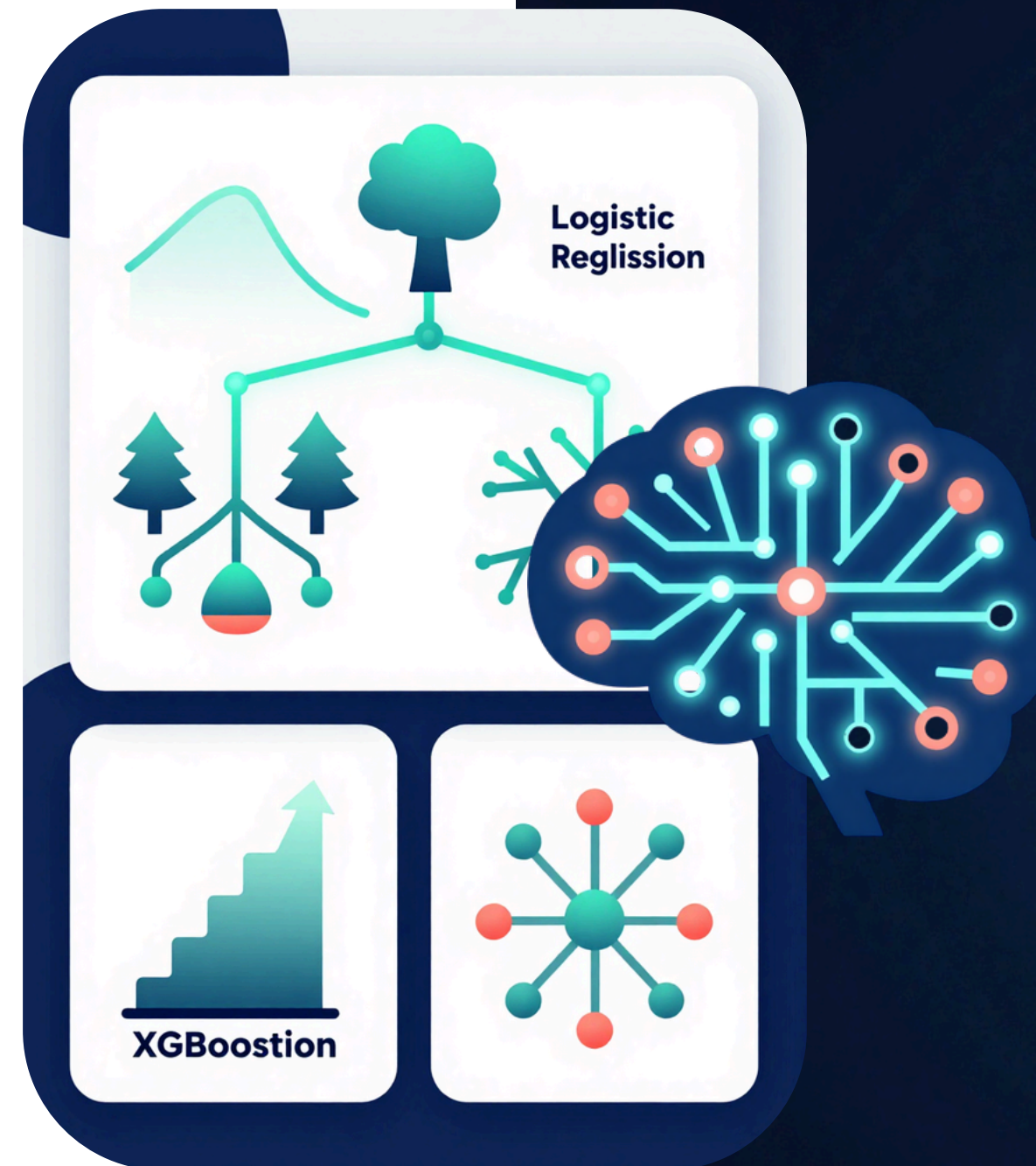
Models Used

Logistic Regression: Solver comparison via cross-validation (lbfgs, liblinear, saga). Class-balanced weighting applied to handle imbalanced classes effectively.

Random Forest: Tuned with randomized search. 500 estimators used with feature importance analysis to identify key clinical predictors.

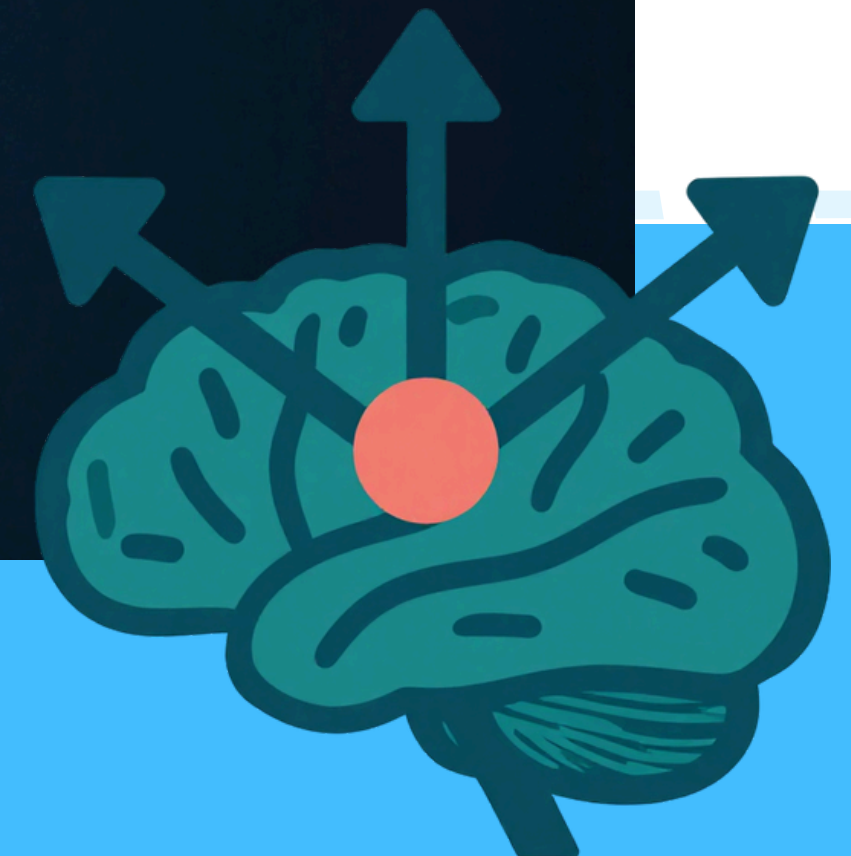
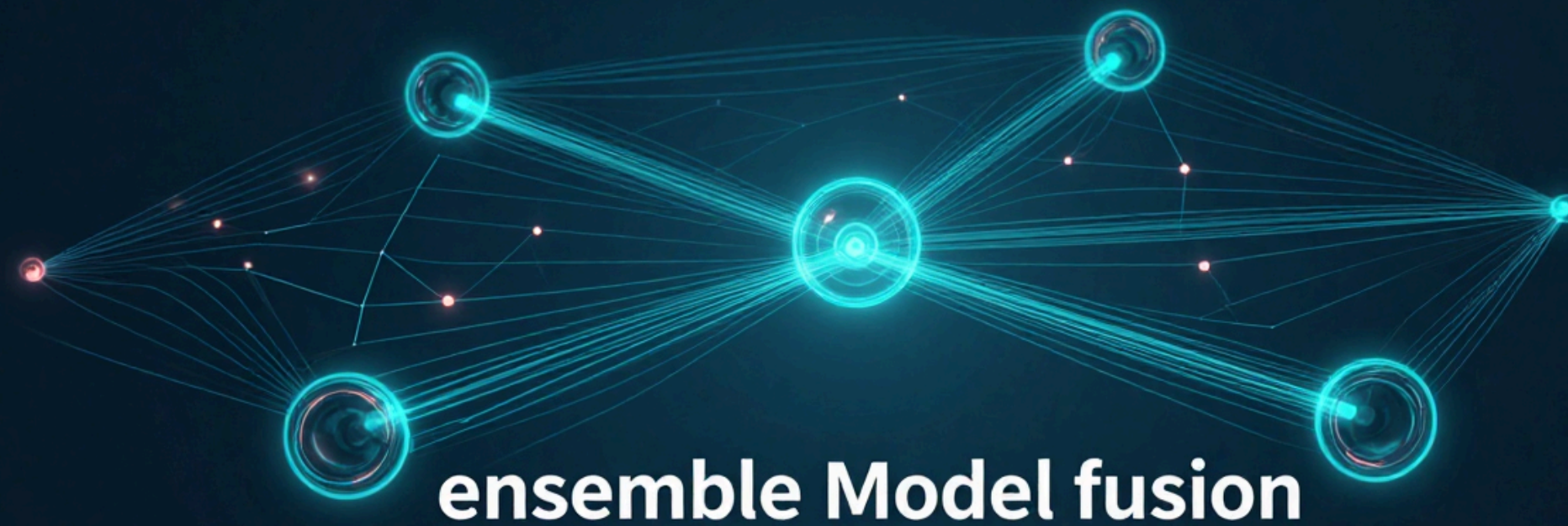
XGBoost: Gradient boosting optimized via randomized search. Feature importance extracted to interpret model decisions.

Deep Learning (MLP): Keras neural network with architecture search. Incorporates Dropout, Batch Normalization, and Early Stopping to prevent overfitting and improve generalization.



→
**Hybrid
Ensemble
Model**

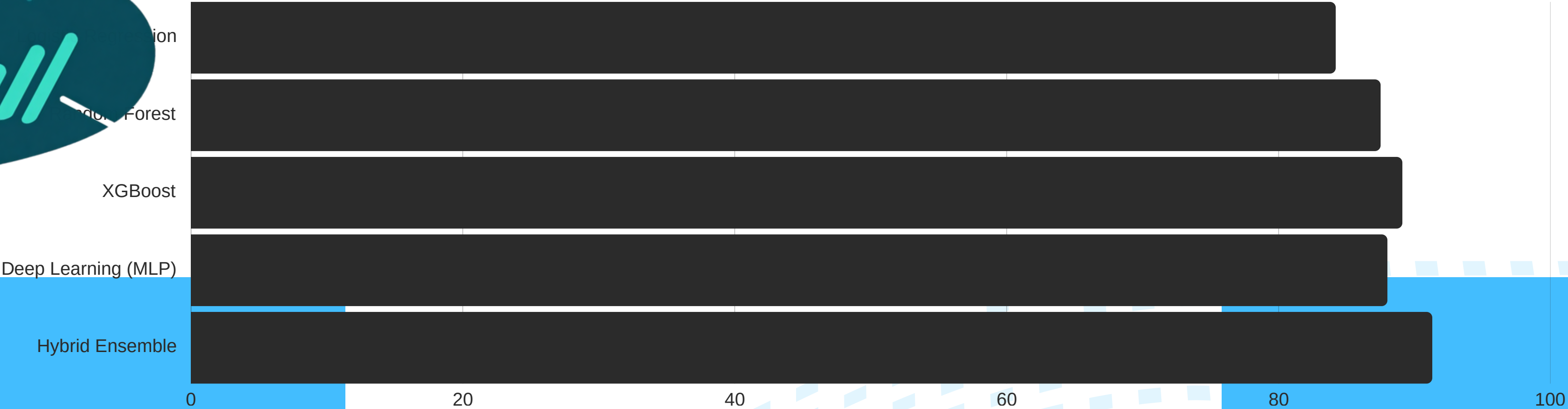
A weighted soft-voting ensemble combines predicted probabilities from all four models: Logistic Regression (15%), Random Forest (25%), XGBoost (35%), and Deep Learning MLP (25%) — delivering robust, high-accuracy final predictions.



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Model Comparison

Test accuracy across all five approaches. The Hybrid Ensemble achieves the highest performance by combining the strengths of Logistic Regression, Random Forest, XGBoost, and Deep Learning.



Source: AI-generated data. Replace or verify before use.

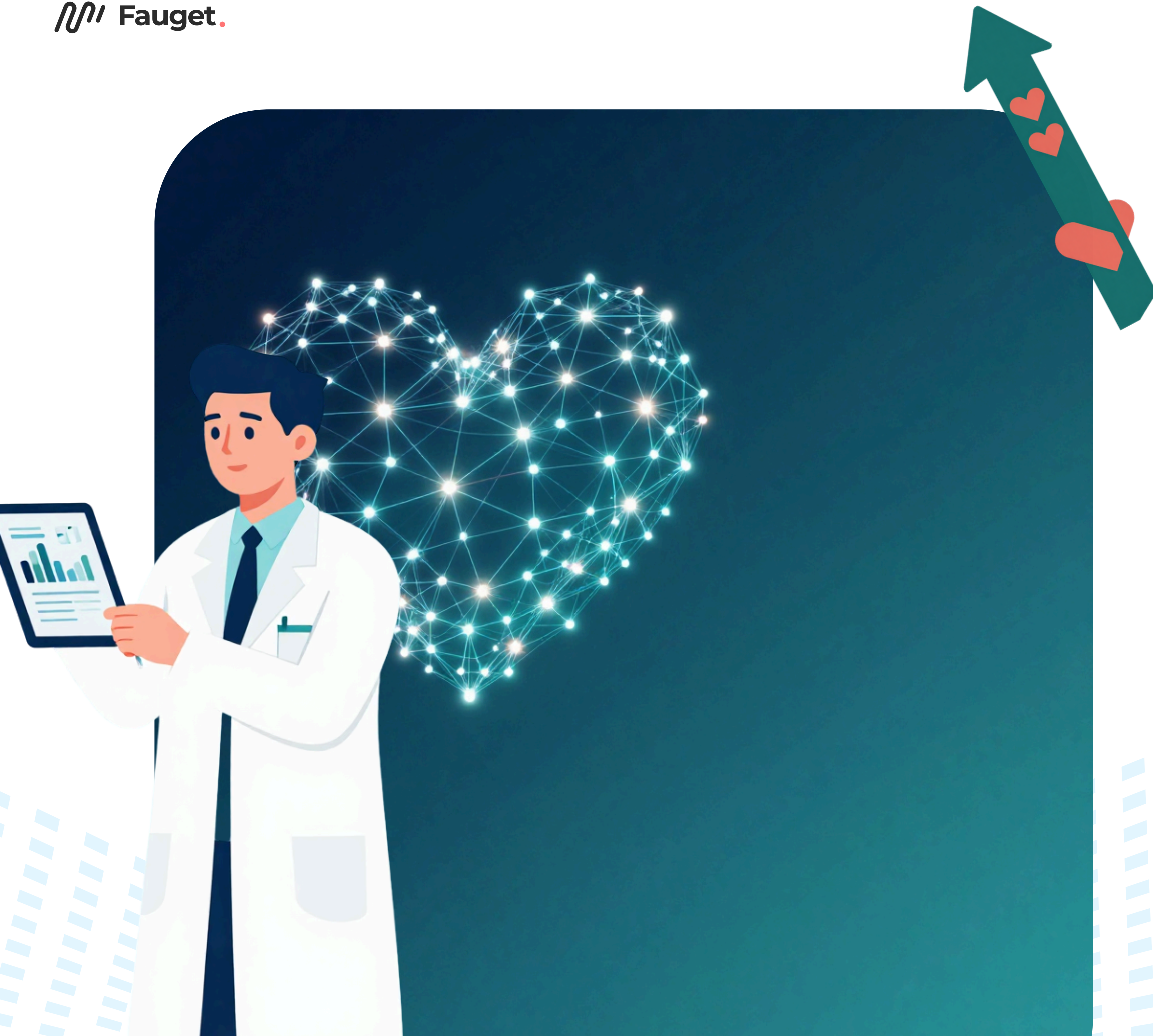


Deployment Web Application

An interactive prediction tool built with the Gradio framework allows users to input patient clinical features — including age, cholesterol, resting BP, ECG results, and more — and receive an instant heart disease prediction.

Users can select from five models: Logistic Regression, Random Forest, XGBoost, Deep Learning (MLP), or the Hybrid Ensemble. Each prediction returns a probability score, supporting both clinical decision-making and research workflows.





→ Conclusion & Future Work

The hybrid ensemble model delivers improved robustness and accuracy for heart disease prediction, supporting early clinical interventions and better patient outcomes.

Future Work:

- Expand dataset size and diversity
- Incorporate SHAP explainability methods
- Deploy as integrated hospital decision support tool

Thank You!

Thank you for your attention. We welcome your questions and invite collaboration on heart disease prediction research.
Contact: presenter@organization.edu

Get In Touch

